NOTE The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83532/6A
30 December 1991
SUPERSEDING
MIL-D-83532/6
10 March 1989

PERFORMANCE SPECIFICATION SHEET

DELAY LINES, ACTIVE,
16-PIN DIP COMPATIBLE,
PROGRAMMABLE 3-BIT, EMITTER-COUPLED LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-D-83532

REQUIREMENTS

Design and construction: Parts shall be of the design, construction, and physical dimensions specified in the appendix of MIL-D-83532. Two case styles (E and F) are available.

Input pulse: Delay lines must be capable of meeting applicable table I and table II requirements with an input pulse having the following characteristics:

A. Leading edge of a positive-going pulse.

B. Minimum pulse width of 50 percent of maximum delay time.

C. Fixed pulse repetition rate equal to ten times the maximum delay time.

D. Duty cycle not to exceed 50 percent.

Schematic: See figure 1.

Delay times: Delay time from input to all taps shall be as specified in table I (+25°C) and table II (-55°C and +125°C).

Output rise time (applied to leading edge only): 7 ns maximum. Measurement conditions (-55°C ≤ T_C ≤ +125°C), V_{EE} = -5.2 V dc, TR_I ≤ 3 ns.

DC parameters (over operating temperature range): See table III.

Truth table: See table IV.

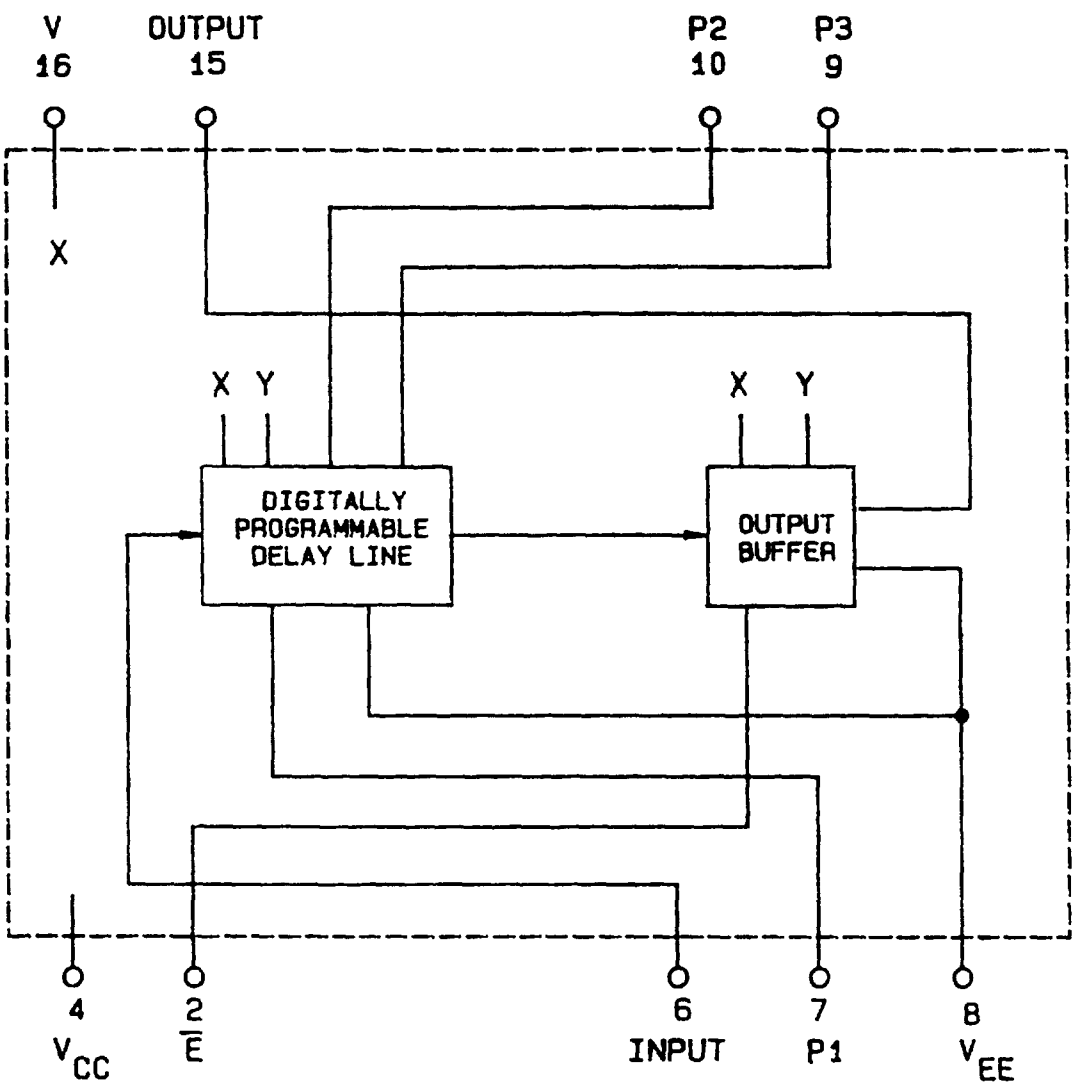
Rated maximum load (fan-out): The delay lines are capable of driving 70 emitter-coupled logic dc loads.

Terminations: Part terminations shall be printed-circuit pin type in accordance with MIL-H-38534.

Operating temperature: Operating temperature range shall be -55°C to +125°.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES
1. Schematic is for general information only
2. Pins 3, 4, 5, 11, 12, 13, and 14 are omitted on level A delay lines
3. All pins are installed on level B delay lines

FIGURE 1. Schematic

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TABLE I Dash numbers and delay characteristics at +25°C, V_{EE} = -5.2 ±.01 volts.

1/ Dash number	2/ 3/ Step zero delay time (ns)	3/ Maximum delay time (ns) (±2 0 ns or 5% whichever is greater)	3/ Delay change per step (ns)	4/ Maximum deviation from programmed delay (ns)
001	3 0 ±2 0	10	1 ±0 4	±1 0 or ±5% (whichever is greater)
002	"	17	2 ±0.6	"
003	"	24	3 ±1 0	"
004	"	31	4 ±1 0	"
005	"	38	5 ±1 5	"
006	"	45	6 ±1 5	"
007	"	52	7 ±1 5	"
008	"	59	8 ±1 5	"
009	"	66	9 ±1 5	"
010	"	73	10 ±1 5	"
011	"	108	15 ±1 5	"
012	"	143	20 ±1 5	"
013	"	178	25 ±1 5	"
014	"	213	30 ±2 0	"
015	"	248	35 ±2 0	"
016	"	283	40 ±2 5	"
017	"	318	45 ±2 5	"
018	"	353	50 ±3 5	"

1/ For PIN, maximum delay time, tabulation and truth table examples, a step zero delay time of 3 ns was assumed

2/ Delay at step zero is referenced to the input pin

3/ The delay is being measured at -1 3 volt level, no load, with the output terminated through a 50 ohm resistor to -2 0 volts

4/ All delay times after step zero are referenced to step zero

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TABLE II Dash numbers and delay characteristics at -55°C and +125°C, V_{EE} = -5 20 ±0 01 volts

1/ Dash number	2/ 3/ Step zero delay time (ns)	3/ Maximum delay time (ns) (±4.0 ns or 8% whichever is greater)	3/ Delay change per step (ns)	4/ Maximum deviation from programmed delay (ns)
001	3 0 ±3 5	10	1 ±0 6	±2 0 or ±8% (whichever is greater)
002	"	17	2 ±0 8	"
003	"	24	3 ±1 0	"
004	"	31	4 ±1 0	"
005	"	38	5 ±1 0	"
006	"	45	6 ±1 2	"
007	"	52	7 ±1 4	"
008	"	59	8 ±1 6	"
009	"	66	9 ±1 8	"
010	"	73	10 ±2 0	"
011	"	108	15 ±2 5	"
012	"	143	20 ±3 0	"
013	"	178	25 ±3 5	"
014	"	213	30 ±4 0	"
015	"	248	35 ±4 0	"
016	"	283	40 ±4 5	"
017	"	318	45 ±5 0	"
018	"	353	50 ±6 0	"

1/ For PIN, maximum delay time, tabulation and truth table examples, a step zero delay time of 3 ns was assumed

2/ Delay at step zero is referenced to the input pin

3/ The delay is being measured at -01 3 volt level, no load, with output terminated through a 50-ohm resistor to -2 0 volts

4/ All delay times after step zero are referenced to step zero

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TABLE III. DC characteristics

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
High level output output voltage	V_{OH}	V_{EE} = -5 2 V V_{IH} = -0 81 V I_{OH} = 425 µA	-1 1	-0 7	V
Low level output voltage	V_{OL}	V_{EE} = -5 2 V V_{IL} = -1 85 V I_{OL} = 0 3 µA	-1 9	-1.615	V
High level input current	I_{IH1}	V_{EE} = -5 2 V, V_{IH} = -0 98 V		25	mA
	I_{IH2}	V_{EE} = -5 2 V, V_{IH} = -0 7 V		13 5	mA
Low level input current	I_{IL1}	V_{EE} = -5.2 V, V_{IL} = -1 825 V		2	mA
	I_{IL2}	V_{EE} = -5.2 V, V_{IL} = -1 63 V	0.3		µA
Low level supply current	I_{ee}	V_{EE} = -5 2 V		-120	mA

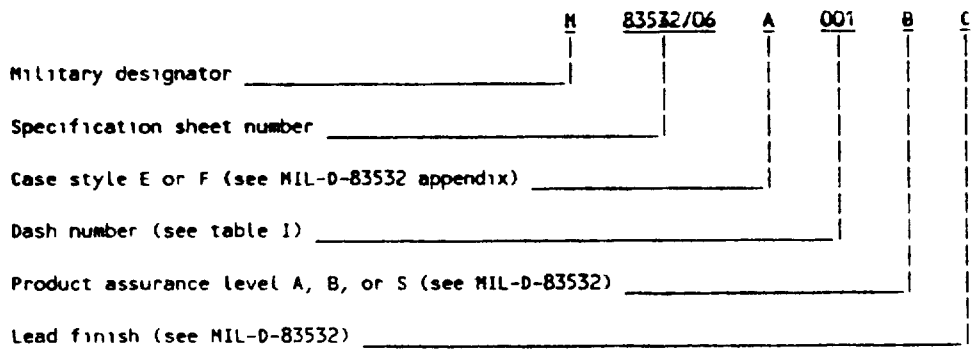
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TABLE IV Truth table.

Programming pins	3 2 1	0 0 0	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
Dash number								
001	3	1	2	3	4	5	6	7
002	3	2	4	6	8	10	12	14
003	3	3	6	9	12	15	18	21
004	3	4	8	12	16	20	24	28
005	3	5	10	15	20	25	30	35
006	3	6	12	18	24	30	36	42
007	3	7	14	21	28	35	42	49
008	3	8	16	24	32	40	48	56
009	3	9	18	27	36	45	54	63
010	3	10	20	30	40	50	60	70
011	3	15	30	45	60	75	90	105
012	3	20	40	60	80	100	120	140
013	3	25	50	75	100	125	150	175
014	3	30	60	90	120	150	180	210
015	3	35	70	105	140	175	210	245
016	3	40	80	120	160	200	240	280
017	3	45	90	135	180	225	270	315
018	3	50	100	150	200	250	300	350

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Part or Identifying Number (PIN) The PIN shall be in the following format



QUALITY ASSURANCE PROVISIONS

Extent of qualification The extent of qualification shall be as specified in MIL-D-83532, for example

Qualification and testing of M83532/06A001A* 1/ and M83532/06A018A* 1/ shall be sufficient to grant qualification to all dash numbers with case style E contained in this specification sheet Similarly, for case style F, qualification and testing of dash numbers 001 and 018 shall be sufficient to grant qualification to all dash numbers in this specification sheet having the same case style as the components tested

Production assurance levels Product assurance level A shall not extend to level B Level B shall not extend to level A

Supersession data MIL-D-83532/6 supersedes DESC drawing number 85078 when a qualified source is available

Revision letters are not used to denote changes due to the extensiveness of the changes

1/ * - Asterisk represents any of the applicable finishes

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Air Force - 11, 19, 99
DLA - ES

User activities
Navy - AS, CG, MC, SH

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5999-0284-06)

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